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Topic 06 : Data Modelling

Part 01 : Data Modelling - Introduction

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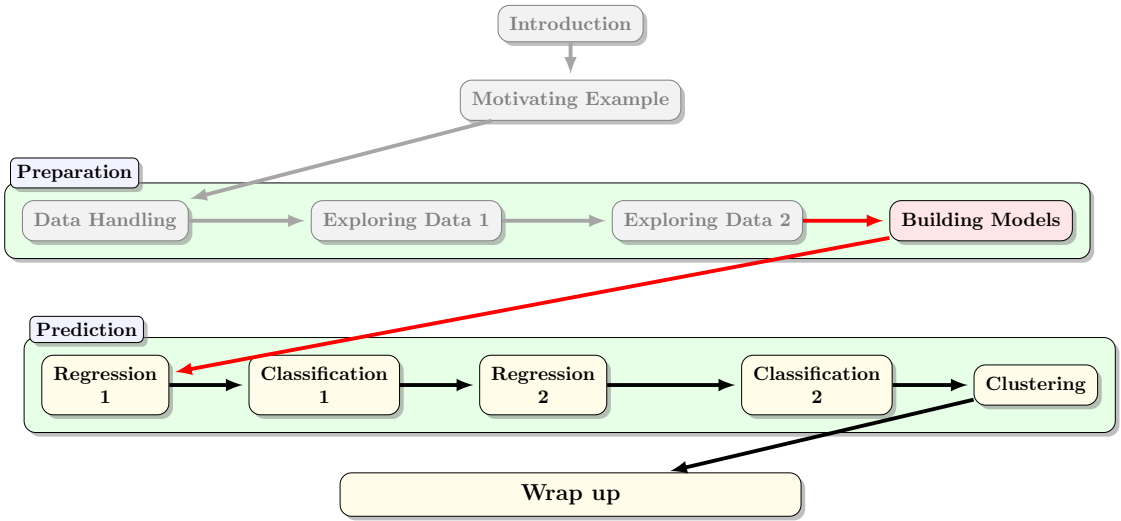
Autumn Semester, 2026

Outline

- Components of a machine learning problem
- Machine learning concepts and notation
- Bias vs variance

Wrap up

AI Engineering (Week 6)



Outline

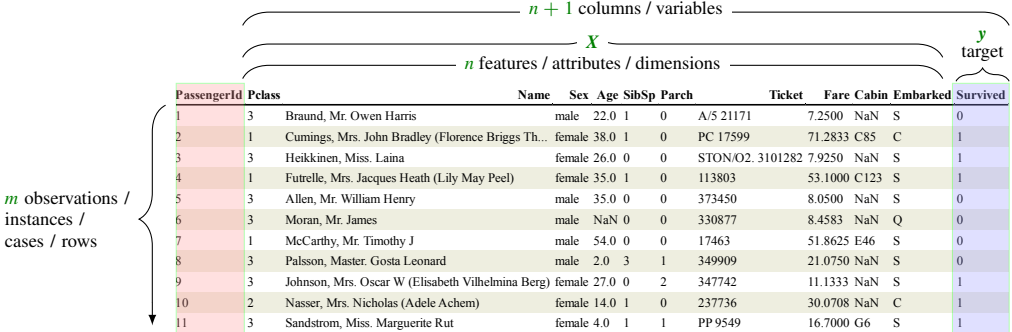
1. Machine Learning (ML) Overview 3
1.1. Review of terminology and notation 4
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1.4. Taxonomy of Machine Learning Methods 11
1.5. Statistical Models vs Machine Learning Models 13

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Terminology / Notation

PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
1	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	0
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7	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S	0
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- A labeled dataset consists of m rows $\times$ $(n + 1)$ columns / variables.
- Use bold to represent vectors and matrices.

Terminology / Notation

$n + 1$ columns / variables

$\boldsymbol{X}$

n features / attributes / dimensions

$\boldsymbol{y}$ target

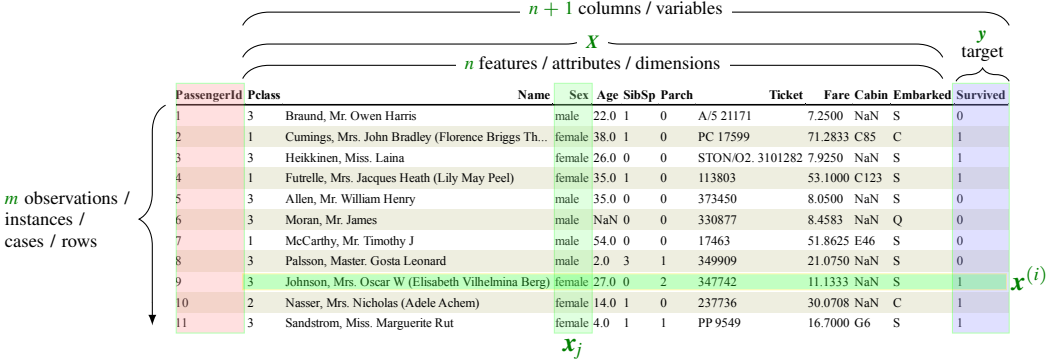
m observations / instances / cases / rows

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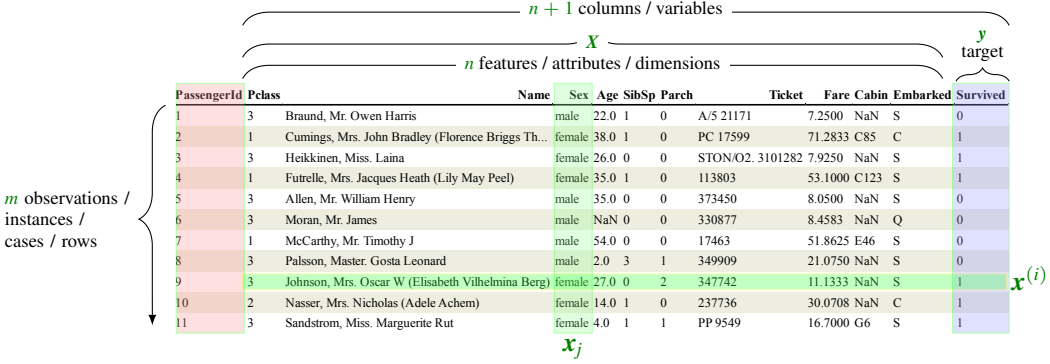
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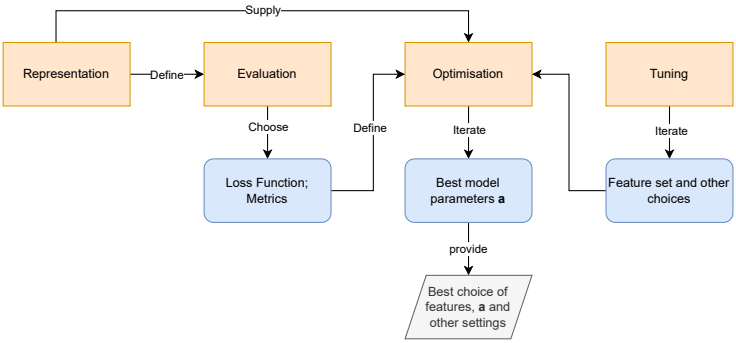
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- So $x_j^{(i)}$ (or $x_{i,j}$) is the i-th observation in the j-th feature ..... $x_j^{(i)}$

Components of a Machine Learning Problem

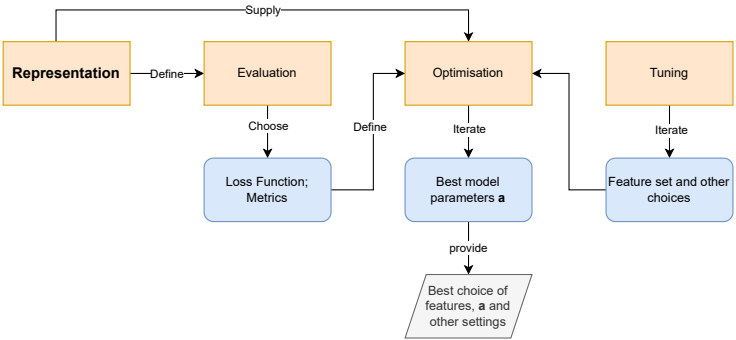
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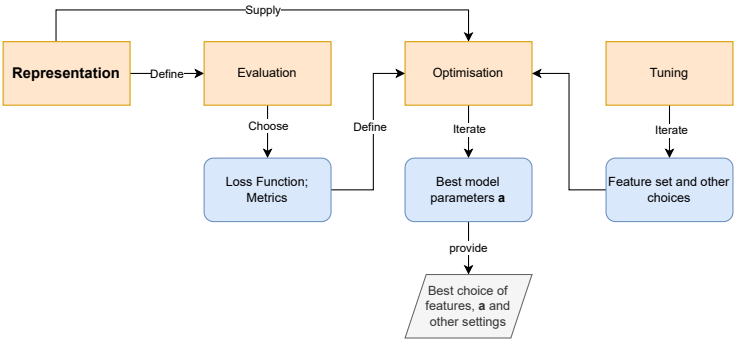
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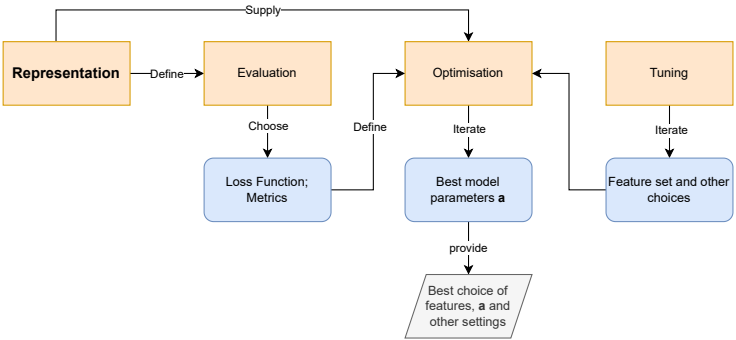


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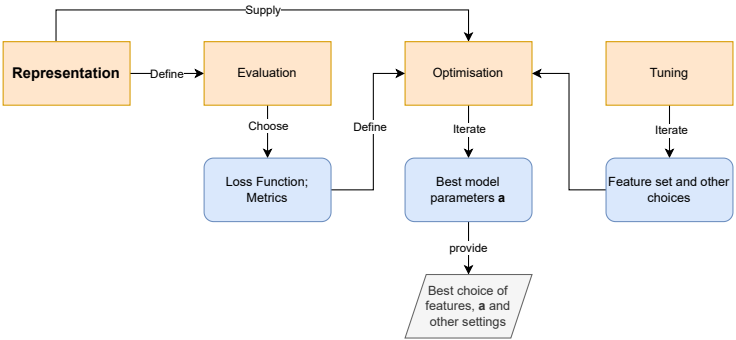
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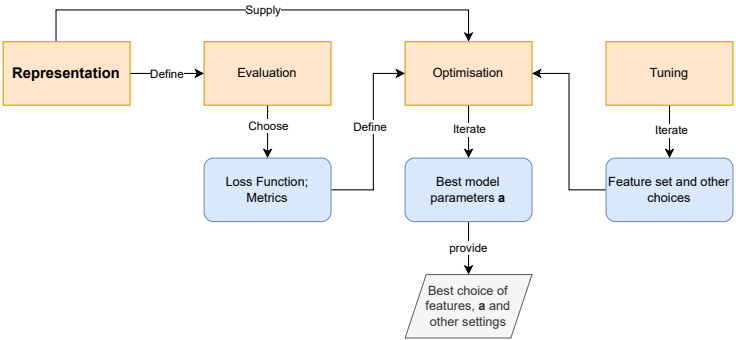
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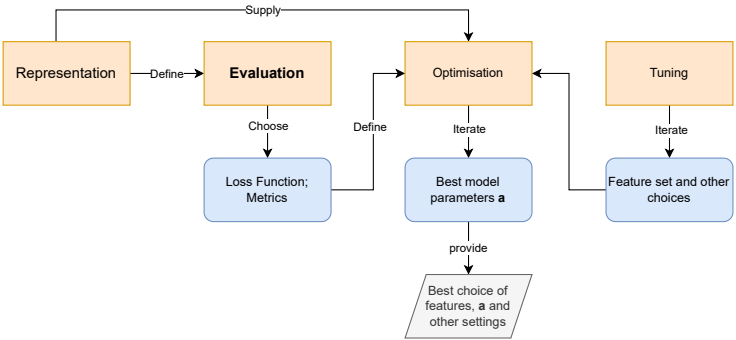
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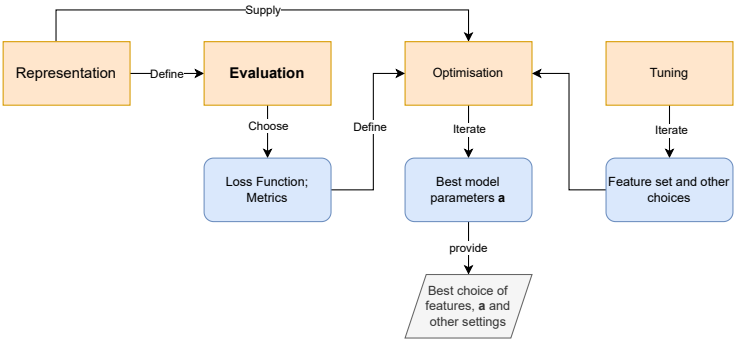


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- These choices are made once - they are foundational for what follows.

Component 2: Evaluation

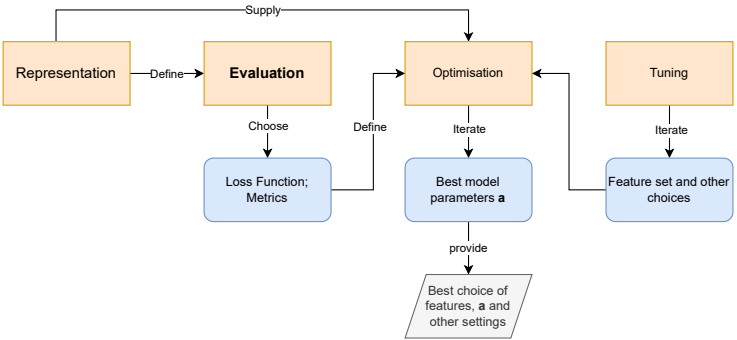


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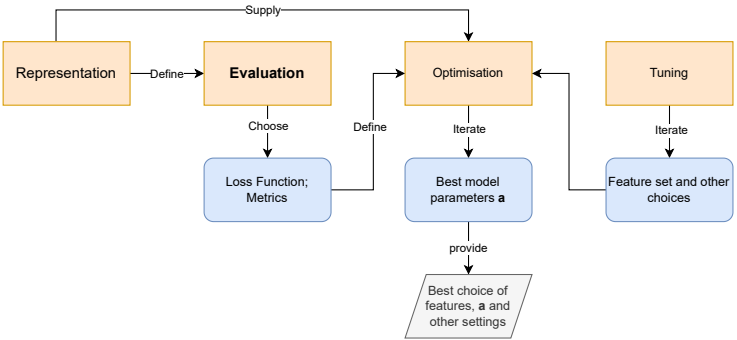
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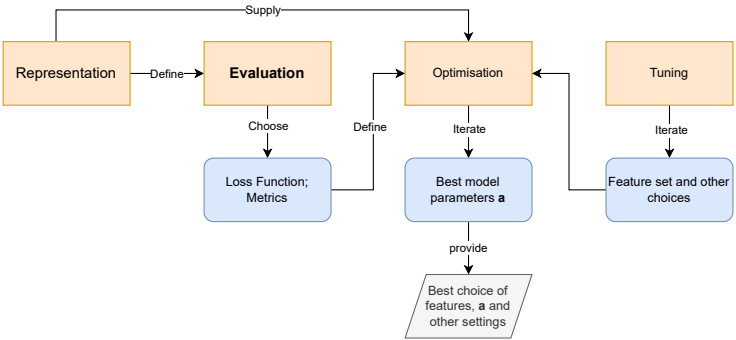
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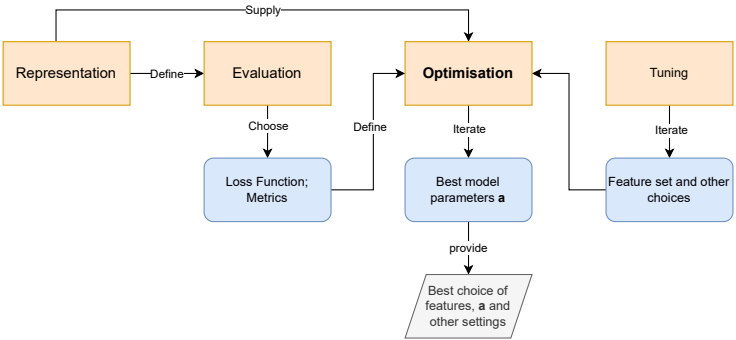
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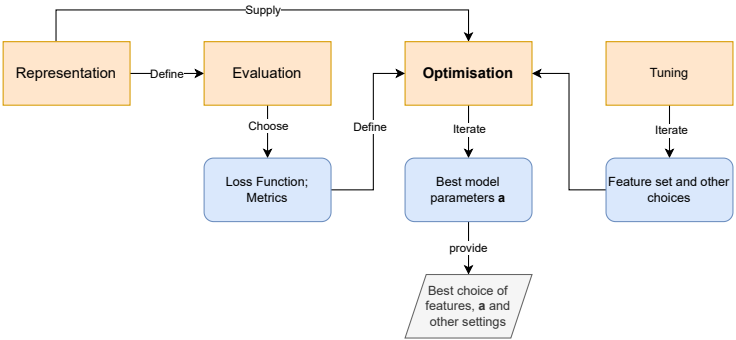


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- For classification: how often does the model predict the right class, so that the predicted target and the actual target match each other?
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- Generally the objective function takes actual and predicted target values and returns a single nonnegative number.

Component 3: Optimisation

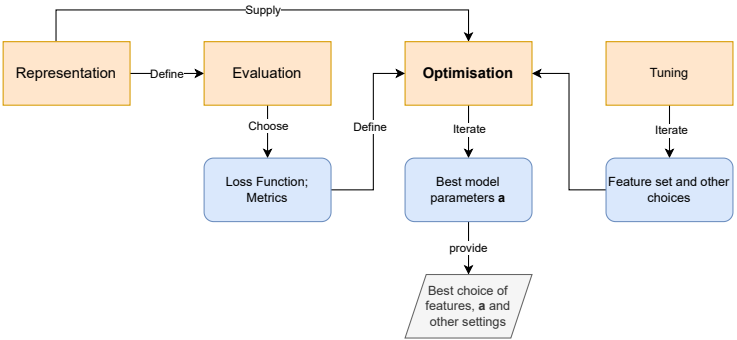


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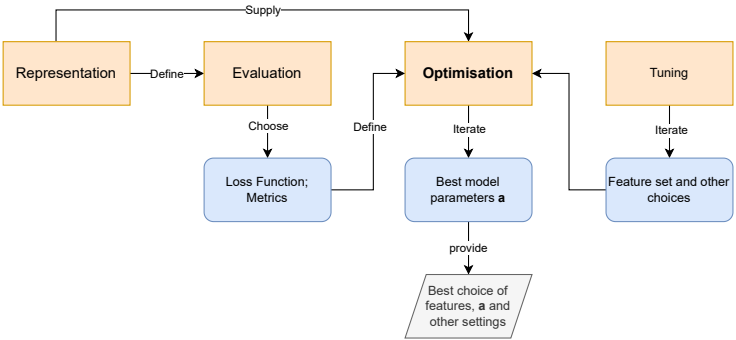
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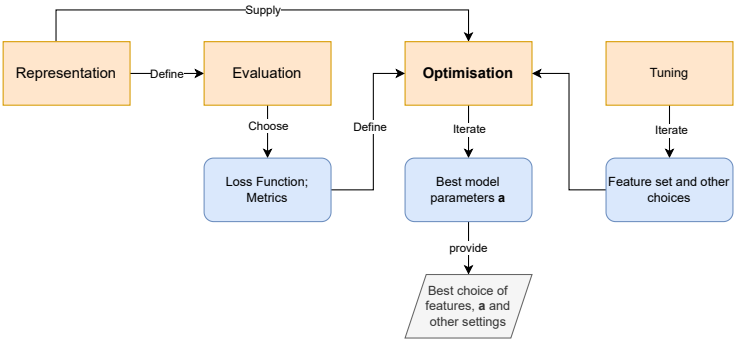
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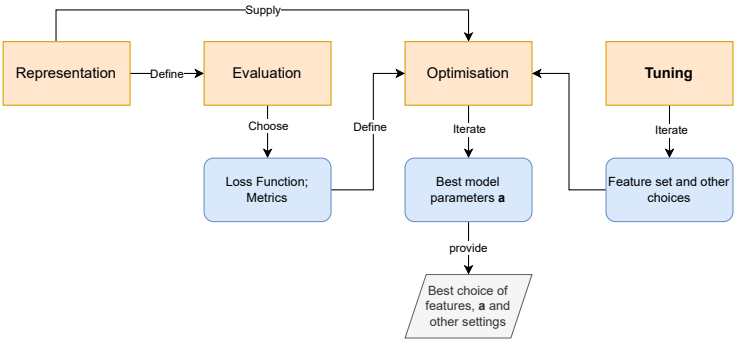
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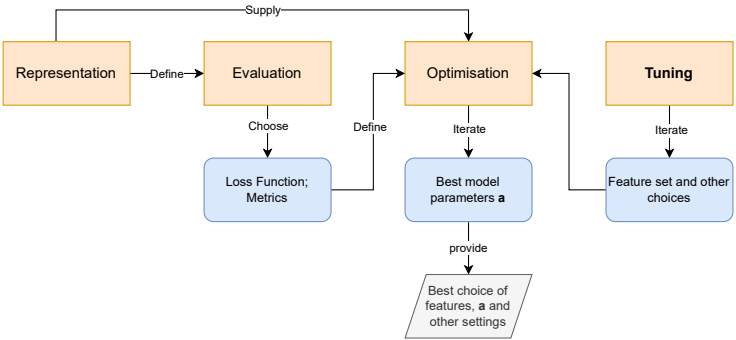


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- The choice of when to terminate the optimisation procedure depends on the Evaluation choice.
- Most of the computational resources are needed for the Optimisation component.

Component 4: Tuning

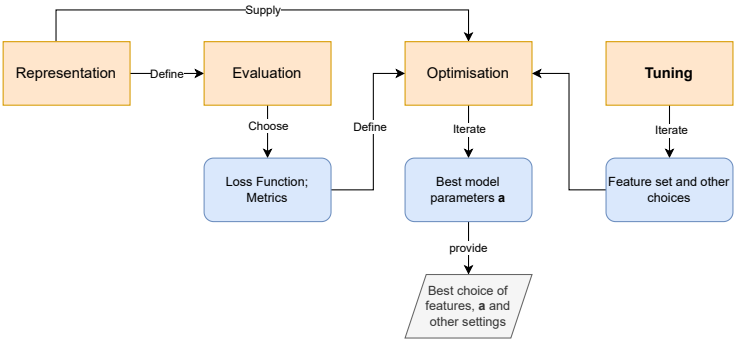


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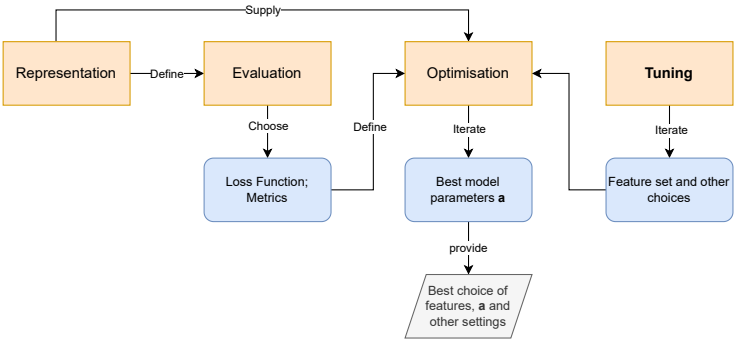
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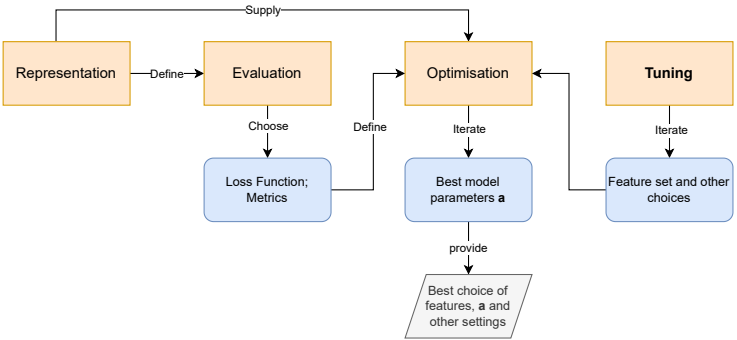
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- Tuning parameters depend on Representation. For example, we have seen one hyperparameter already: k in the KNN classifier.
- For Tuning, GridSearch (exhaustive over a range) and RandomSearch (pick represntative values) are popular ways to search for the best hyperparameter values.

Data Modelling (aka Machine Learning)

As alternative to the four component (Representation / Evaluation / Optimisationi/ Tuning) viewpoint we can think of a machine learning problem as

Definition 1 (Machine Learning)

Study of algorithms that improve their performance P at some task T with experience E.

Well defined learning task: $< P, T, E >$

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- How complex is the task?
- Task type: classification, regression, ...
- Linear vs nonlinear?
- What family of functions should be used?

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- Linear vs nonlinear?
- What family of functions should be used?

- How many historical observations are needed?
- How accurate/noisy is the data?
- Do we have missing values?
- Is the data representative?

Taxonomy of Machine Learning Models . . .

. . . by Intuition/Motivation

. . . by Algorithmic Properties

. . . by Fixed/Variable Number of Parameters

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- Geometric models use intuitions from geometry such as separating (hyper-)planes, linear transformations and distance metrics.
- Probabilistic models view learning as a process of reducing uncertainty, modelled by means of probability distributions.
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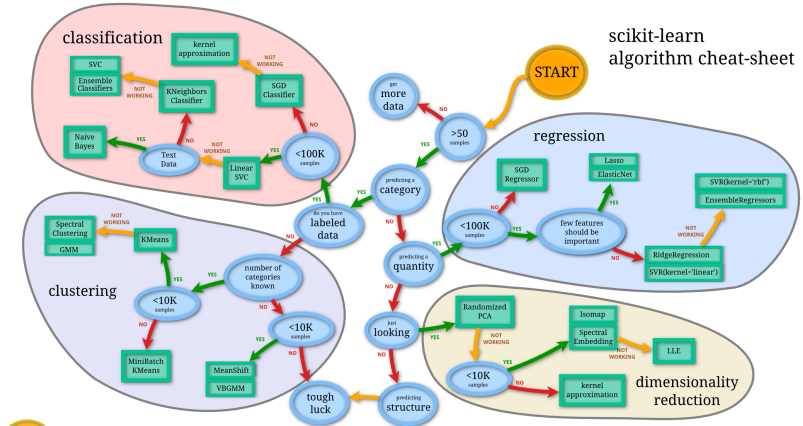
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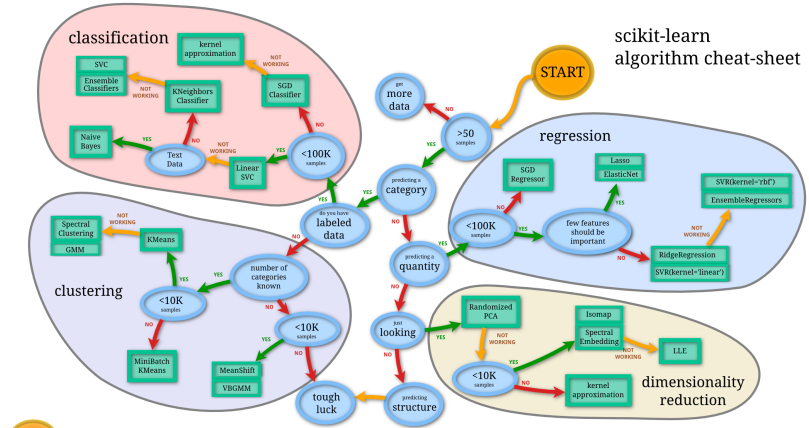
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- Parametric models have a fixed number of parameters.
- In non-parametric models the number of parameters grows with the amount of training data.

Aside: Scikit-learn Flowchart of Models (Shallow Learners)

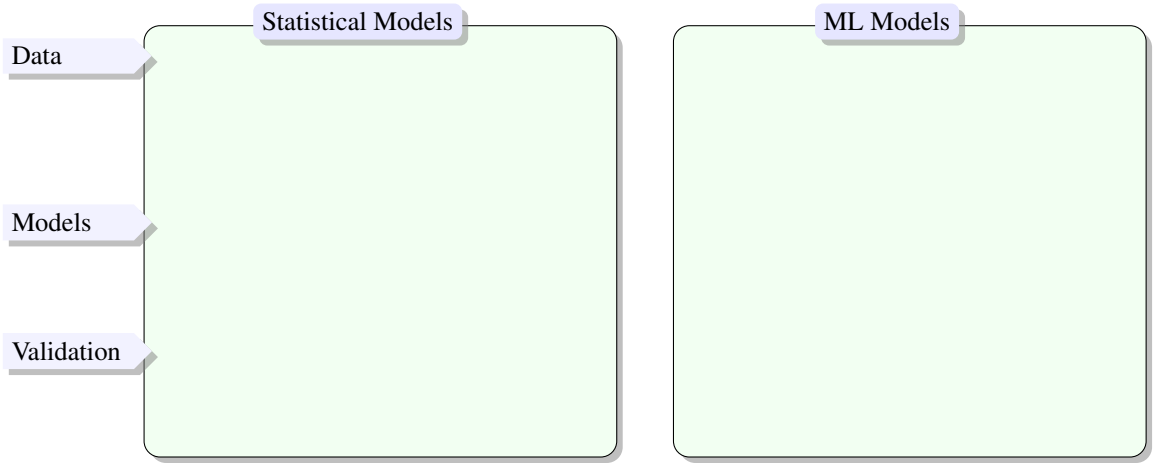


Aside: Scikit-learn Flowchart of Models (Shallow Learners)



A neural network with more than one hidden layer is called a deep learner, all other learners are shallow learners.

Statistical Models vs Machine Learning Models



Statistical Models vs Machine Learning Models

	Statistical Models	ML Models
Data	• Usually small (< 1000 observations) • Low dimension (< 10 variables) • Can have detailed understanding of data • Data is clean — human has looked at each data point	• Can be huge (million+ observations) • Large dimension (1000+, more for vision) • Too large for human to parse / understand • Data not clean — humans can't afford to understand/fix each point
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Splitting data into train+test(+validation) is vital

Outline

1. Machine Learning (ML) Overview 3
1.1. Review of terminology and notation 4
1.2. Components of a Machine Learning Problem 5
1.3. Problem–Task–Experience Perspective 10
1.4. Taxonomy of Machine Learning Methods 11
1.5. Statistical Models vs Machine Learning Models 13

2. Modelling Process 14
2.1. Models and error 16

The Pipeline Metaphor

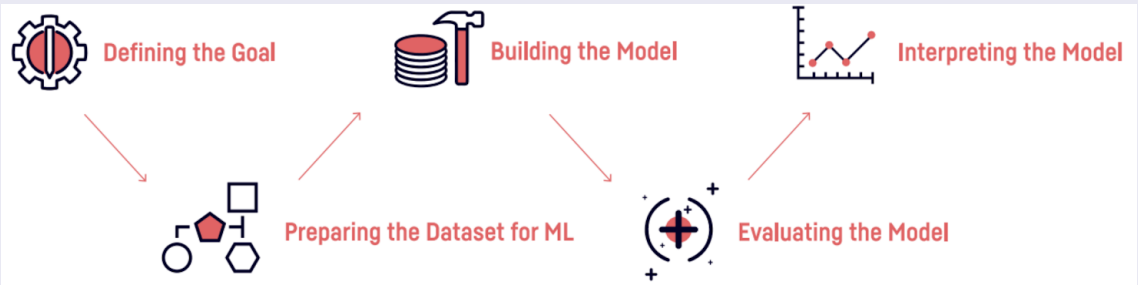
Model Building Pipeline



Source: Dataiku

The Pipeline Metaphor

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Comments

- We saw the first two stages in previous weeks
- This week we look at the remaining stages
- Of course this pipeline is a simplification. In reality it is iterative.

What does a (supervised learning) model look like?

Definition 2 (Linear Model)

General form of linear model used in this module looks like

$$y_i \sim f_i^{(1)} + f_i^{(2)} + \ldots + f_i^{(\mathrm{n})}$$

where y_i is the value of the response variable (target) for row (observation) i, and $f_i^{(\mathrm{j})}; j = 1, \ldots, n$ is the value of the jth feature for that observation.

In words: each target value y_i in the data is modelled as a linear combination of the model parameters $\boldsymbol{a}$ and the features, plus some error.

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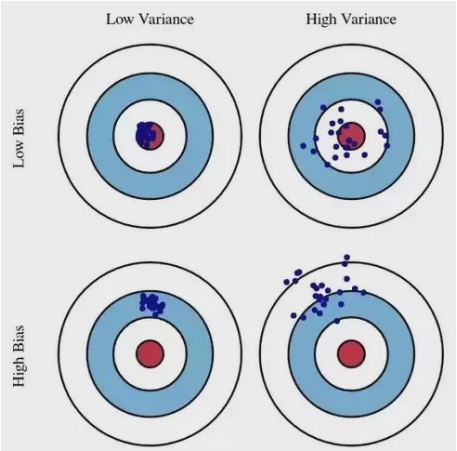
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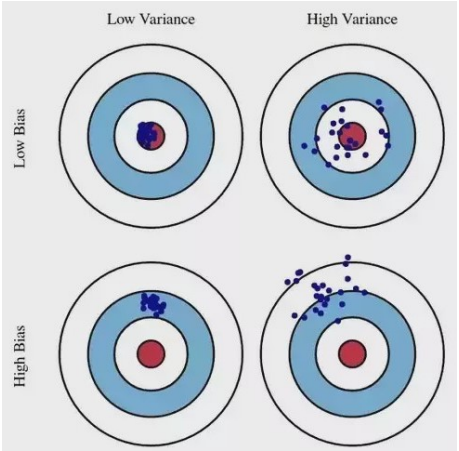
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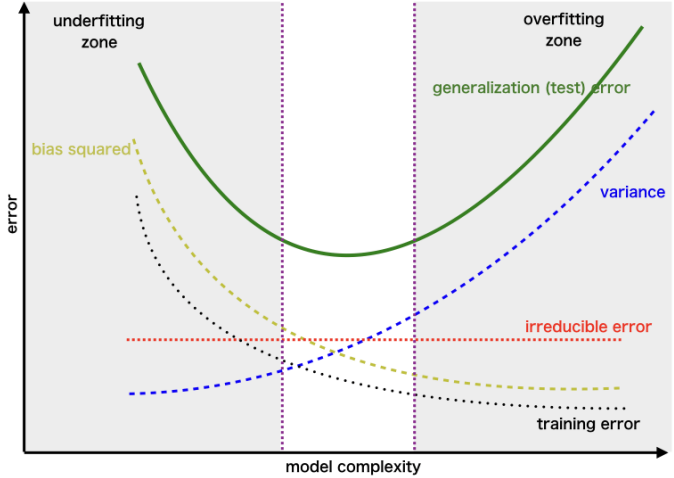
The goal of modelling is to find a so that the *prediction error* (loss function $\sim \|\varepsilon\|$) is a minimum.

Bias-Variance and Total Error

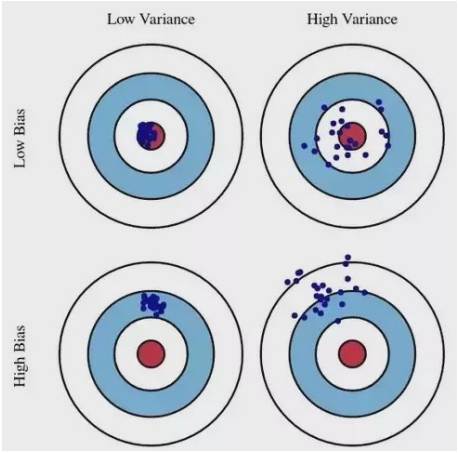


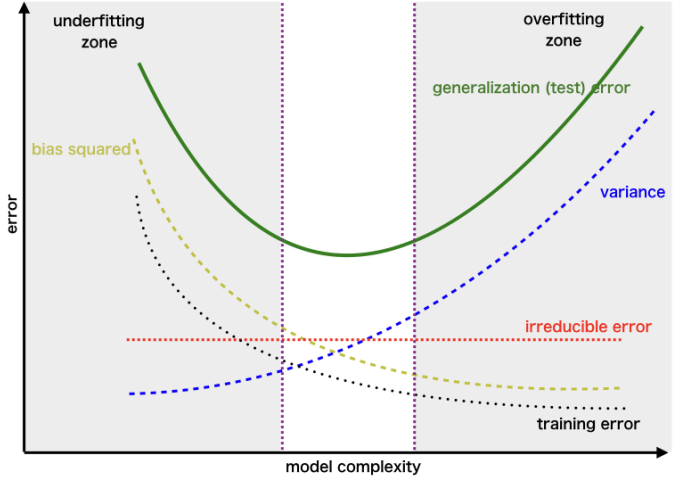
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Bias-Variance and Total Error





Look for parameters ***a*** that minimise the generalization error (estimated using the test set that was not used during training)

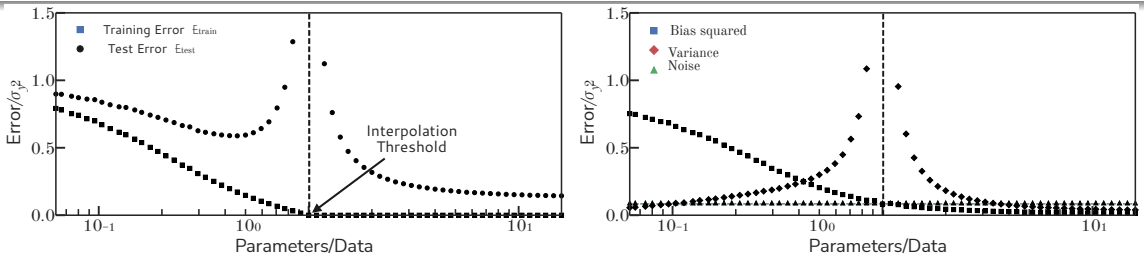
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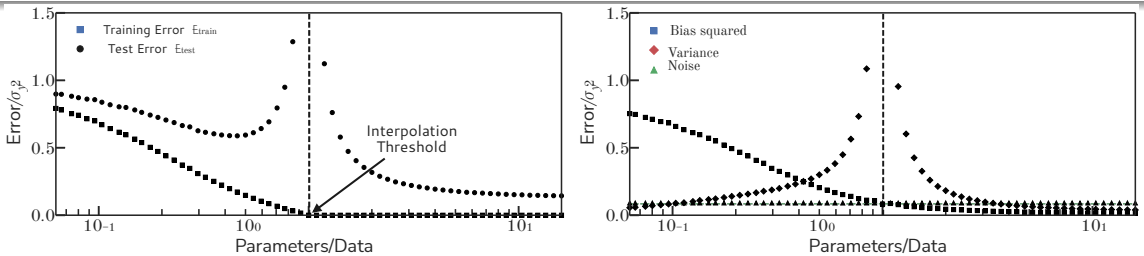
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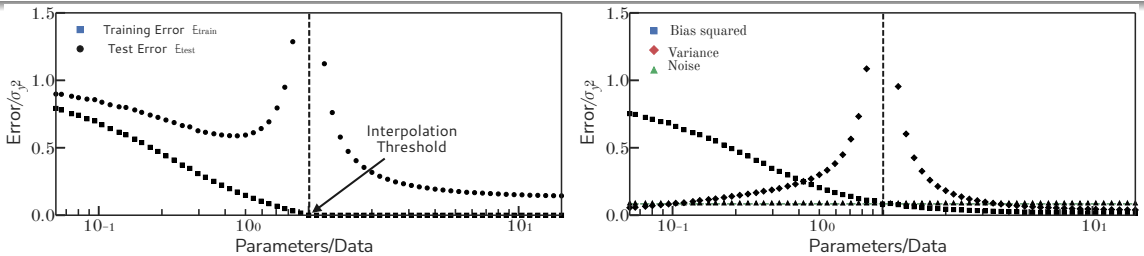
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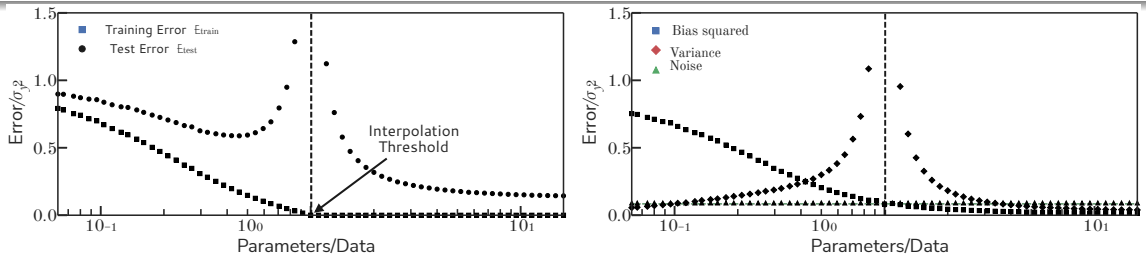
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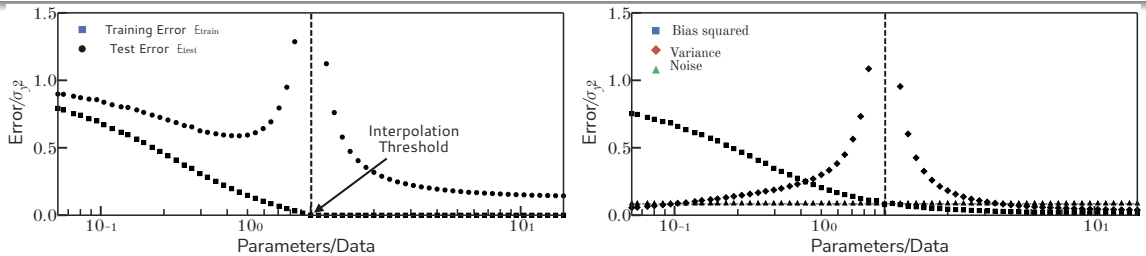
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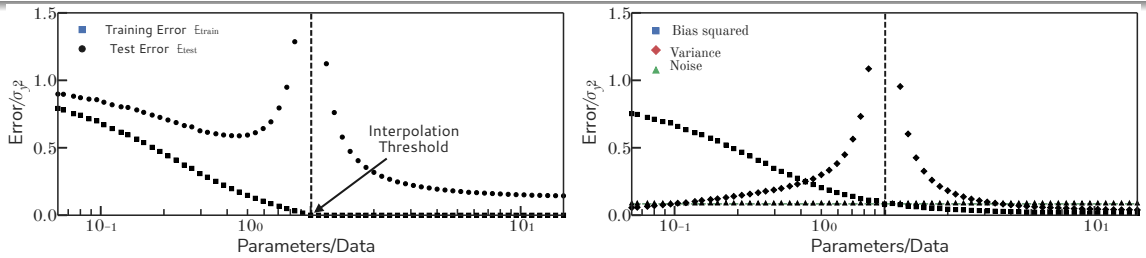
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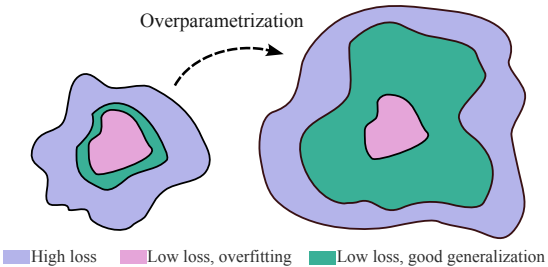
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- For deep learning - see AI Engineering 2 (semester 2)

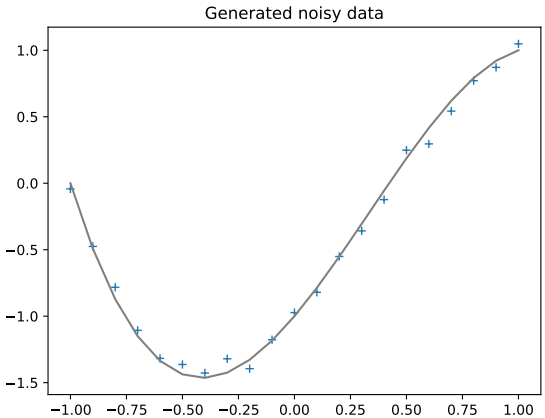
Effects of Double Descent



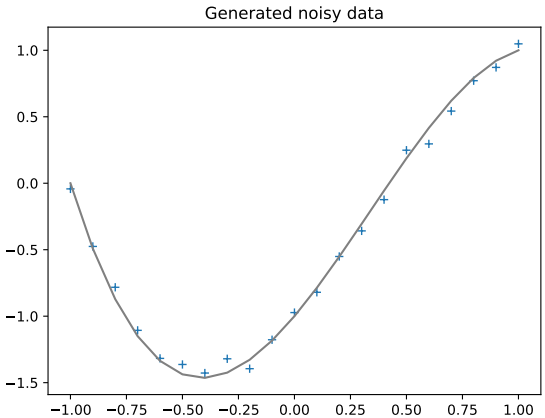
Extending to high dimensions

- Conventional models are usually not over parametrised: the number of features (columns) is usually much less than the number of cases (rows), i.e., the feature matrix is long and narrow
- Since double descent does not occur, the middle optimal settings zone (indicated by green colour adjoining) is narrow
- With deep learning, overparameterised models are easier to set up, so double descent allows a much wider optimal settings zone.
- This also allows for smaller bias without dramatically larger variance.

Bias-Variance Example: Noisy data



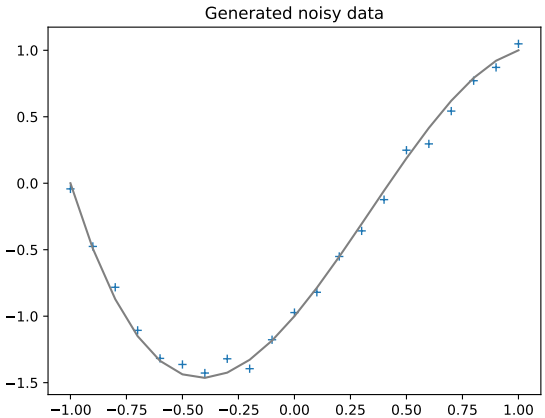
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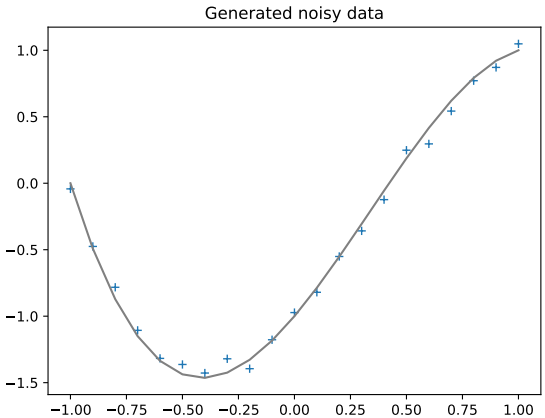
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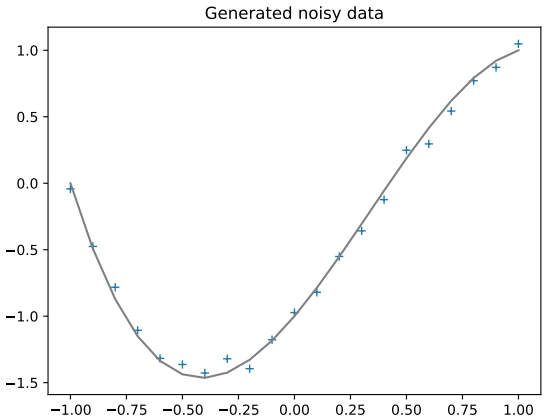
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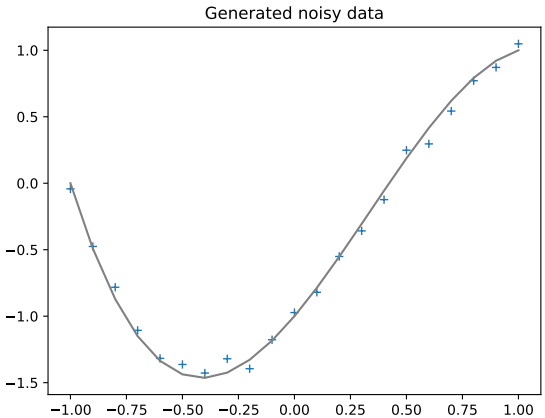
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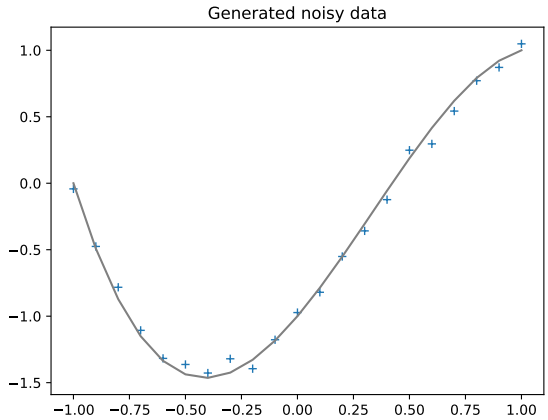
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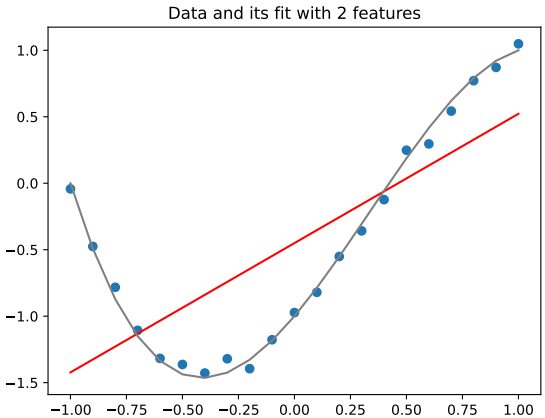


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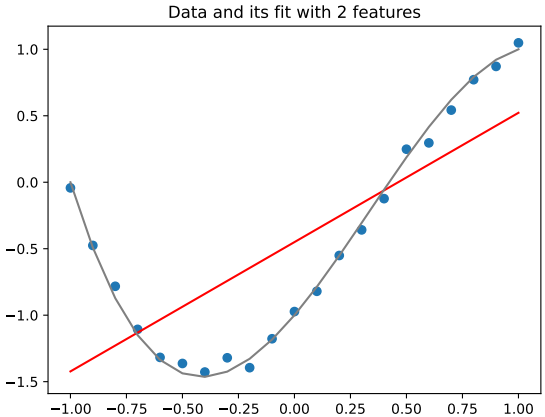
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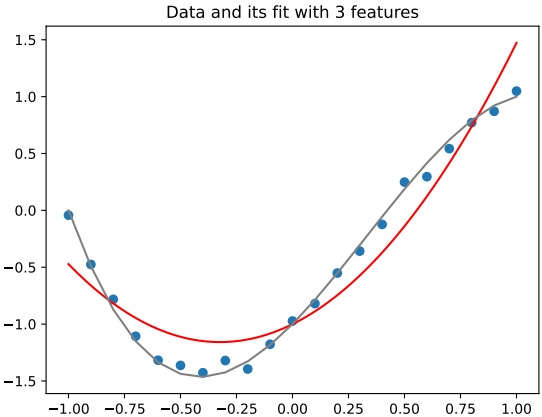
Look for the number of features that minimise the loss function

High Bias, Low variance

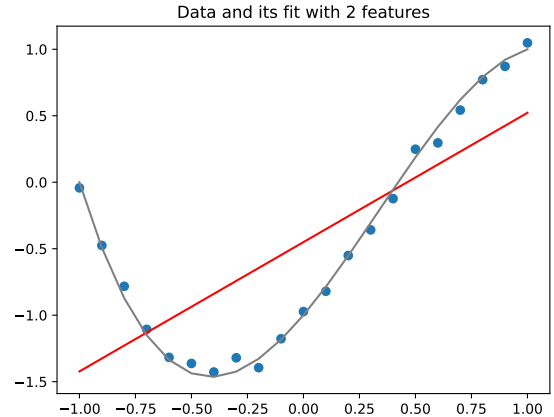


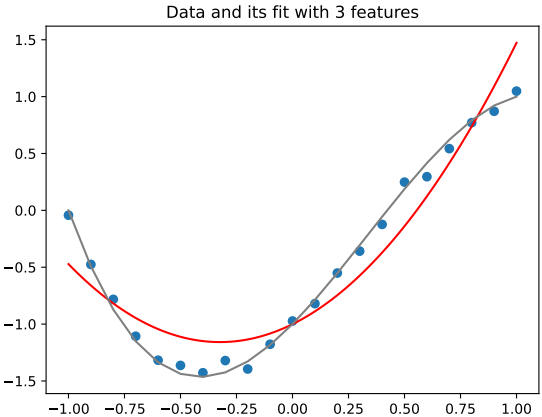
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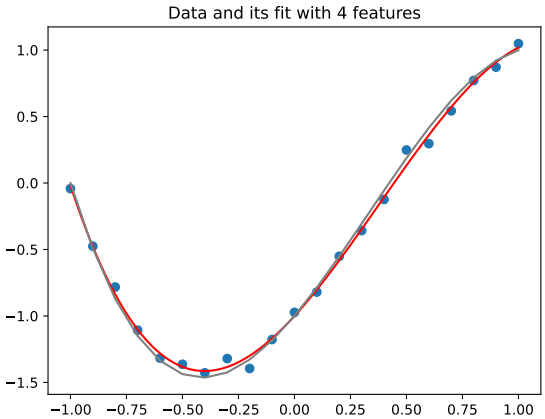
High Bias, Low variance



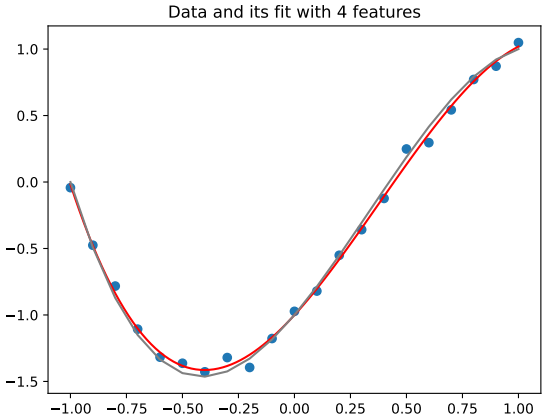


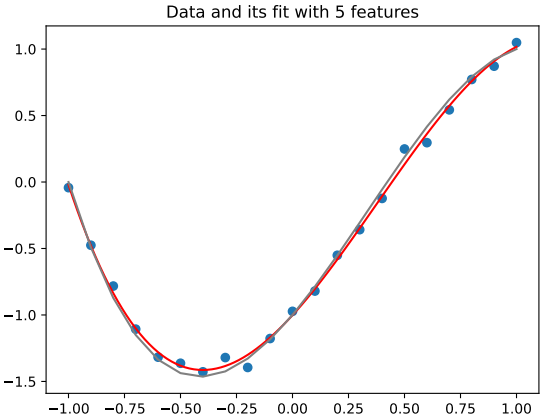
Need more features: add more - but which ones?...

Low Bias, Low variance

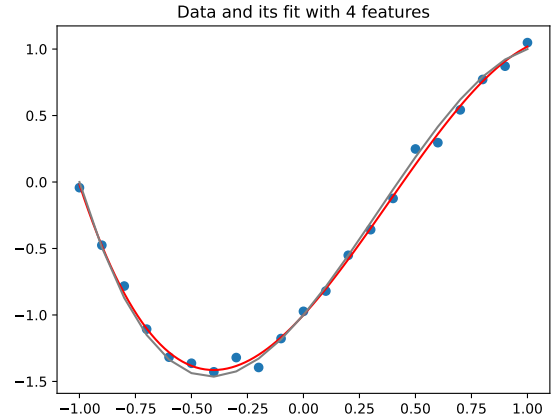


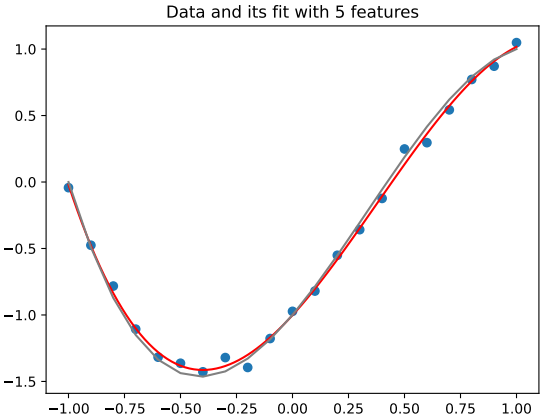
Low Bias, Low variance





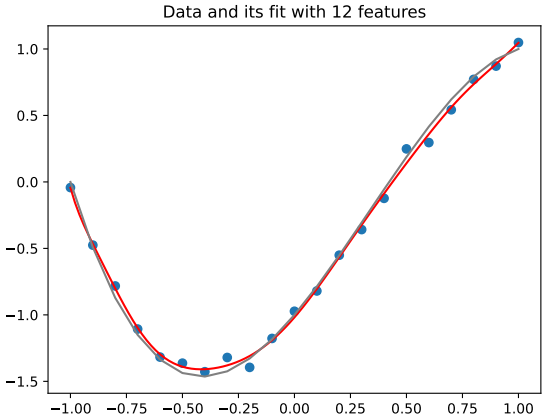
Low Bias, Low variance



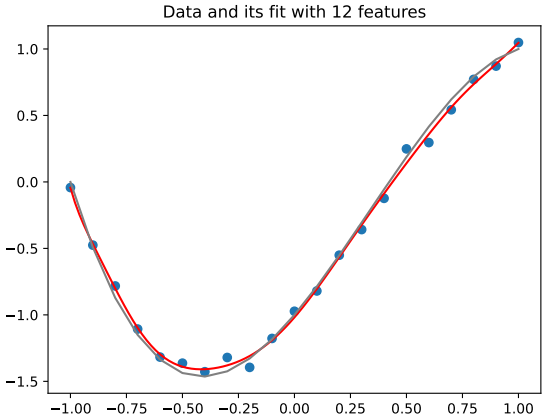


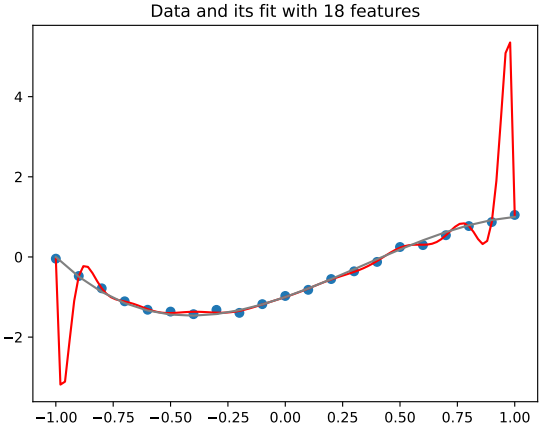
About the right number of features…

Low Bias, High variance

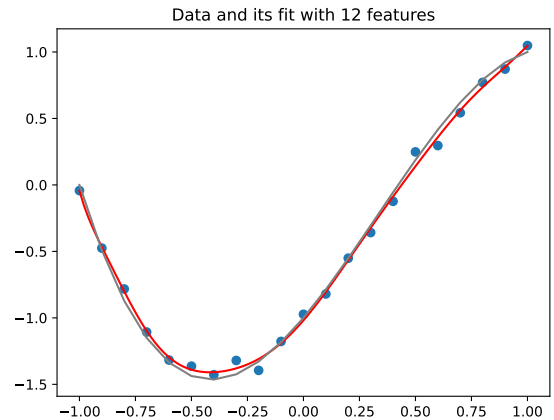


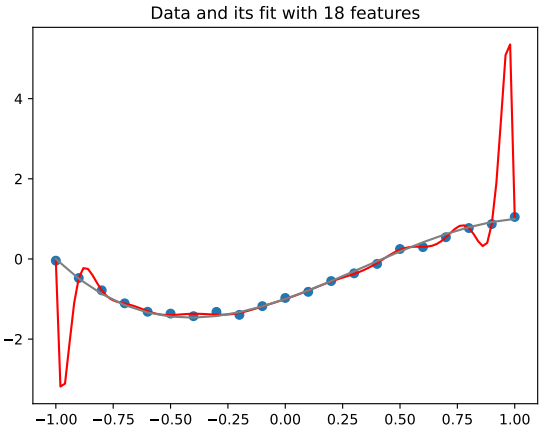
Low Bias, High variance





Low Bias, High variance





Too many features: remove some, but which ones?...

Example Model Types

Model	Applications	Concerns
Logistic Regression	X-ray classification	Regression with transformed variable
Fully connected networks	Classification	Classical ANN: choose encoding and size
Convolutional Neural Networks	Image processing	deep learning - choose segmentation
Recurrent Neural Networks	Voice recognition	ANN with feedback - how much?
Random Forest	Fraud Detection	Ensemble method - how many?
Reinforcement Learning	Learning by trial and error	Choose goal and penalties
Generative Models	Text, Image creation	Choose parameters
K-means	Segmentation	Choose distance function and k
k-Nearest Neighbors	Recommendation systems	Choose distance function and k
Bayesian Classifiers	Spam and noise filtering	Deal with imbalances

Before you start...

Does a *pre-trained* model exist?

Before you start...

Does a *pre-trained* model exist?

Transfer Learning

- Building a model from scratch is resource-intensive
- Open source data and model exist, particularly for deep learning (not in this nmodule)
- Most frameworks provide example models that can be used as a template
  - Select a similar model
  - Prune it (remove unnecessary terms)
  - Train using the pruned model as a starting point

When model training is very expensive, such as training a LLM, foundation models provide a good starting point. Hugging Face provides a wide selection of such models.